

Impact of 6G Network Performance on Manufacturing Efficiency in Smart Factories

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Abstract—The emergence of Industry 4.0 and Industry 5.0 manufacturing environments has significantly increased the dependence of industrial operations on ultra-reliable and low-latency communication networks. In modern smart factories, production systems, predictive maintenance platforms, industrial automation controllers, and AI-driven decision engines continuously exchange large volumes of real-time data across connected manufacturing infrastructures. While machine failures and process inefficiencies have traditionally been considered the primary causes of productivity degradation, communication network instability can independently reduce manufacturing efficiency even when mechanical systems operate normally. Despite the growing adoption of next-generation 6G communication technologies, manufacturing organizations often lack analytical frameworks capable of quantifying the impact of network latency and packet loss on operational performance, production quality, and manufacturing efficiency. This paper presents an Impact of 6G Network Performance on Manufacturing Efficiency in Smart Factories analysis framework. The objective of this study is to investigate how network latency, packet loss, and communication reliability influence production speed, quality control outcomes, operational error rates, and overall manufacturing efficiency. The dataset consists of industrial machine telemetry records containing machine operating conditions, network performance indicators, production metrics, quality measurements, and efficiency classifications. Multiple analytical techniques and manufacturing-specific Key Performance Indicators (KPIs) were developed, including Network Stability Index, Latency Sensitivity Score, Packet Loss Impact Ratio, and Network-Efficiency Correlation Metrics. An interactive Streamlit dashboard was developed to visualize network behavior, manufacturing efficiency trends, latency diagnostics, packet loss impacts, quality degradation patterns, and operational risk zones. The analysis demonstrates that communication network performance serves as a critical and independent factor affecting manufacturing efficiency in 6G-enabled smart factories. The proposed framework provides actionable intelligence for identifying network-related production bottlenecks, optimizing industrial communication infrastructures, improving operational reliability, and supporting data-driven decision-making for next-generation smart manufacturing environments.

Index Terms—Smart Manufacturing, 6G Networks, Manufacturing Efficiency, Network Latency Analysis, Packet Loss Analysis, Industry 4.0, Industry 5.0, Industrial Analytics, Streamlit Dashboard, Predictive Manufacturing, Industrial IoT, Network Stability Index

I. INTRODUCTION

The rapid evolution of Industry 4.0 and the emergence of Industry 5.0 have transformed traditional manufacturing environments into highly connected, intelligent, and data-driven production ecosystems. Modern smart factories inte-

grate advanced communication technologies, Industrial Internet of Things (IIoT) devices, autonomous production systems, artificial intelligence (AI), predictive maintenance platforms, and cyber-physical systems to enable real-time monitoring and automated decision-making. These technologies continuously generate and exchange large volumes of operational data across interconnected manufacturing infrastructures, creating new opportunities for improving productivity, efficiency, product quality, and operational reliability. A fundamental requirement of smart manufacturing systems is the availability of reliable and ultra-low latency communication networks. Manufacturing equipment, robotic systems, quality-control stations, digital twins, and AI-driven optimization engines depend on continuous data exchange to coordinate production activities and support real-time decision-making. As industrial environments become increasingly connected, communication networks are evolving from supportive infrastructure components into mission-critical operational assets. Any degradation in communication performance may directly influence manufacturing outcomes, even when machines themselves remain mechanically healthy. The development of sixth-generation (6G) communication technologies is expected to further accelerate the transformation of smart manufacturing environments. Compared to previous wireless communication generations, 6G networks promise significantly lower latency, enhanced reliability, ultra-high data rates, intelligent network slicing capabilities, and seamless machine-to-machine communication. These characteristics position 6G as a key enabling technology for next-generation industrial automation systems. Through network slicing and intelligent resource allocation, 6G infrastructures can support diverse manufacturing workloads while maintaining stringent quality-of-service requirements across production facilities. Despite these technological advancements, many manufacturing organizations continue to face challenges in understanding the operational consequences of communication network instability. Production slowdowns, increased defect rates, declining efficiency levels, and elevated error rates are frequently attributed to equipment malfunction, maintenance issues, or process inefficiencies. However, communication-related factors such as network latency variation and packet loss may independently contribute to operational degradation without triggering obvious mechanical failure indicators. As a result, organizations often lack visibility into the true relationship between network performance and manufacturing efficiency. The growing complexity of modern

industrial systems has increased the need for analytical frameworks capable of evaluating communication network behavior alongside traditional manufacturing metrics. Network latency can delay machine coordination, interrupt real-time decision execution, and reduce production responsiveness. Similarly, packet loss can compromise communication reliability, increase operational errors, and negatively affect quality-control processes. Understanding how these network conditions influence production performance is therefore essential for designing resilient smart manufacturing infrastructures and optimizing industrial communication investments. The availability of large-scale manufacturing telemetry data provides new opportunities for investigating these relationships. Modern industrial systems continuously record machine operating conditions, production metrics, network performance indicators, quality measurements, and maintenance-related information. By integrating network analytics with manufacturing performance analysis, organizations can identify operational vulnerabilities, quantify communication-related risks, and establish evidence-based network optimization strategies. However, many existing manufacturing analytics approaches primarily focus on machine health, predictive maintenance, and process optimization while providing limited visibility into communication network effects.

This paper addresses these challenges by developing an Impact of 6G Network Performance on Manufacturing Efficiency in Smart Factories analytical framework. The proposed system investigates how network latency and packet loss influence production speed, quality-control defect rates, operational error rates, and overall manufacturing efficiency. Multiple manufacturing-specific Key Performance Indicators (KPIs) are developed, including Network Stability Index, Latency Sensitivity Score, Packet Loss Impact Ratio, and Network-Efficiency Correlation metrics. An interactive Streamlit dashboard is also developed to visualize network performance behavior, efficiency trends, communication reliability patterns, and manufacturing risk zones. The proposed framework provides actionable intelligence for understanding the role of communication infrastructure in industrial operations and supports data-driven decision-making for next-generation smart manufacturing environments. By quantifying the relationship between network performance and manufacturing outcomes, the framework enables organizations to identify communication bottlenecks, improve operational resilience, optimize network investments, and enhance overall production efficiency within 6G-enabled smart factories.

II. LITERATURE REVIEW

The rapid advancement of Industry 4.0 technologies has transformed traditional manufacturing environments into highly connected and intelligent production systems. Modern smart factories increasingly rely on Industrial Internet of Things (IIoT) devices, cyber-physical systems, artificial intelligence, and advanced communication infrastructures to support real-time monitoring and autonomous decision-making. As manufacturing systems become more dependent on contin-

uous connectivity, communication network performance has emerged as a critical factor influencing operational efficiency and production reliability.

A first group of studies focuses on Industry 4.0 architectures and smart manufacturing ecosystems. Zhou et al. [1] discussed the technological foundations of Industry 4.0 and highlighted the importance of intelligent manufacturing systems, automation, and industrial connectivity. Similarly, Lasi et al. [2] examined the evolution of smart factories and emphasized the integration of digital technologies across manufacturing processes. These studies established the conceptual foundations of modern industrial systems but provided limited quantitative analysis regarding how communication network performance directly affects manufacturing efficiency.

A second group of research investigates Industrial Internet of Things (IIoT) environments and industrial communication infrastructures. Xu et al. [3] explored the role of IIoT technologies in enabling data-driven manufacturing and real-time industrial monitoring. Lu [4] further analyzed smart manufacturing architectures and highlighted the importance of connected industrial devices for operational optimization. Although these studies emphasize connectivity as a fundamental requirement, they primarily focus on system architecture and device integration rather than evaluating the operational impact of network degradation on production performance.

Another important research direction examines wireless communication technologies for industrial environments. Saad et al. [5] discussed the evolution toward beyond-5G and 6G communication systems and identified ultra-low latency, enhanced reliability, and intelligent networking as key requirements for future industrial applications. Tariq et al. [6] further explored 6G-enabled industrial communication frameworks and highlighted the potential benefits of network slicing, intelligent resource allocation, and machine-to-machine communication. While these studies provide valuable insights into future communication infrastructures, they largely focus on technological capabilities rather than measuring how latency and packet loss influence manufacturing outcomes.

Several studies have also investigated predictive maintenance, machine health monitoring, and manufacturing performance optimization. Lee et al. [7] proposed data-driven predictive maintenance frameworks for industrial environments and demonstrated the value of sensor analytics for reducing downtime. Carvalho et al. [8] examined manufacturing performance indicators and operational efficiency metrics for smart factories. However, these studies predominantly focus on machine condition monitoring, equipment failures, and maintenance optimization while providing limited visibility into communication-related performance degradation.

Recent research has additionally explored cyber-physical manufacturing systems and industrial analytics. Monostori et al. [9] investigated cyber-physical production systems and highlighted the increasing importance of real-time communication for autonomous manufacturing operations. Similarly, Tao et al. [10] examined digital twin technologies and their dependence on reliable industrial data exchange. Although

these studies acknowledge the importance of communication infrastructures, they rarely quantify the relationship between network instability and production efficiency using operational manufacturing data.

Based on the literature, several important research gaps can be identified. First, most existing studies focus on Industry 4.0 architectures, IIoT frameworks, predictive maintenance systems, or communication technologies independently without integrating network performance analytics with manufacturing efficiency analysis. Second, limited research quantitatively evaluates how network latency and packet loss influence production speed, quality-control defect rates, operational errors, and efficiency classifications within industrial environments. Third, existing works generally emphasize communication technology design and network architecture development rather than investigating communication performance as an independent driver of manufacturing outcomes. Finally, few studies develop practical decision-support platforms capable of combining network diagnostics, manufacturing performance evaluation, KPI generation, and visual analytics within a unified framework.

To address these gaps, this paper develops an Impact of 6G Network Performance on Manufacturing Efficiency in Smart Factories analytical framework. Unlike previous studies that primarily focus on industrial architectures, machine health monitoring, or communication technologies in isolation, the proposed system directly investigates the relationship between network latency, packet loss, production speed, quality degradation, operational errors, and manufacturing efficiency. The framework introduces manufacturing-specific Key Performance Indicators (KPIs), including Network Stability Index, Latency Sensitivity Score, Packet Loss Impact Ratio, and Network-Efficiency Correlation metrics. Furthermore, an interactive Streamlit dashboard is developed to provide real-time visualization of network behavior, operational performance, communication risk zones, and optimization opportunities. By integrating network analytics and manufacturing performance analysis within a single decision-support environment, the proposed framework provides actionable intelligence for designing resilient, high-performance 6G-enabled smart factories.

III. PROBLEM STATEMENT

Modern smart factories generate massive volumes of operational, machine, and communication data on a continuous basis. Industrial environments equipped with 6G communication infrastructure, Industrial Internet of Things (IIoT) devices, cyber-physical systems, and intelligent automation platforms rely heavily on real-time data exchange to support manufacturing operations. Production equipment, quality-control systems, predictive maintenance engines, and autonomous decision-making platforms continuously communicate through networked industrial environments to maintain operational efficiency and production reliability. Despite the increasing deployment of advanced communication technologies, many manufacturing organizations still lack analytical frameworks capable of determining how network performance influences

operational outcomes. Manufacturing inefficiencies are often attributed to machine failures, maintenance issues, equipment degradation, or process bottlenecks. However, communication-related factors such as network latency variation and packet loss may independently contribute to declining production performance even when machines remain mechanically healthy. Within 6G-enabled smart factory environments, communication delays can disrupt machine coordination, delay control decisions, reduce production responsiveness, and negatively affect operational throughput. Similarly, packet loss can compromise data integrity, increase operational errors, reduce communication reliability, and contribute to higher product defect rates. These communication-related effects may remain hidden within large-scale manufacturing datasets, making it difficult for organizations to distinguish network-induced inefficiencies from traditional equipment-related problems.

Despite having access to machine telemetry data and network performance measurements, manufacturing organizations face several analytical challenges, including:

- Lack of visibility into how network latency affects manufacturing efficiency.
- Difficulty identifying latency thresholds that trigger production slowdowns.
- Limited understanding of how packet loss influences operational error rates.
- Inability to quantify the relationship between communication reliability and product quality.
- Limited insight into which efficiency levels are most sensitive to network instability.
- Difficulty distinguishing network-related performance degradation from machine-related failures.
- Absence of standardized metrics for evaluating communication-driven manufacturing risks.
- Lack of a centralized analytical platform for monitoring network performance and manufacturing efficiency simultaneously.

As a result, organizations may misallocate network infrastructure investments, misdiagnose production issues, and overlook communication bottlenecks that negatively affect operational performance. Without quantitative evidence regarding the impact of network behavior on manufacturing outcomes, decision-makers face challenges in designing resilient and high-performance industrial communication systems. This paper aims to develop an analytical framework capable of evaluating the impact of 6G network performance on manufacturing efficiency using machine telemetry, production metrics, quality-control indicators, and communication network measurements. The proposed system investigates the influence of latency and packet loss on production speed, operational errors, defect rates, and efficiency classifications while providing actionable insights through an interactive Streamlit dashboard. The framework supports data-driven decision-making and enables a deeper understanding of communication-driven performance behavior within next-generation smart manufacturing environments.

IV. OBJECTIVES

The primary objective of this paper is to develop a manufacturing analytics framework capable of evaluating the impact of 6G network performance on operational efficiency, production quality, and manufacturing reliability within smart factory environments.

The specific objectives are:

- 1) To analyze industrial machine telemetry data and identify network performance trends across manufacturing operations.
- 2) To evaluate network latency and packet loss distributions and classify communication conditions into different network quality levels.
- 3) To investigate the relationship between network performance indicators and manufacturing efficiency classifications.
- 4) To quantify the impact of network latency on production speed and identify latency thresholds associated with operational slowdowns.
- 5) To assess the influence of packet loss on operational error rates and quality-control defect rates.
- 6) To examine how communication network behavior affects manufacturing performance under different machine operation modes.
- 7) To develop manufacturing-specific Key Performance Indicators (KPIs), including Network Stability Index, Latency Sensitivity Score, Packet Loss Impact Ratio, and Network-Efficiency Correlation metrics.
- 8) To identify communication reliability thresholds and network risk zones that may negatively affect production performance.
- 9) To develop an interactive Streamlit dashboard for visualizing network performance, manufacturing efficiency, quality-control outcomes, and operational risk indicators.
- 10) To generate actionable recommendations that support network optimization, operational resilience, production reliability, and data-driven decision-making in 6G-enabled smart factories.

V. DATASET DESCRIPTION

The dataset used in this study represents operational and communication records collected from a simulated 6G-enabled smart manufacturing environment. The dataset captures machine operating conditions, network performance indicators, production metrics, quality-control measurements, and efficiency classifications required for analyzing the relationship between communication network behavior and manufacturing performance. The collected data provides a comprehensive view of both industrial operations and communication reliability, enabling the investigation of how network latency and packet loss influence production efficiency, product quality, and operational stability.

The dataset contains approximately 100,000 manufacturing records collected from multiple industrial machines operating under different production conditions. Each record

represents a machine observation containing information related to machine operating mode, environmental conditions, power utilization, communication performance, production output, quality-control measurements, predictive maintenance readiness, and manufacturing efficiency status. The dataset provides sufficient diversity for evaluating network-induced performance degradation across various operational scenarios. Unlike traditional manufacturing datasets that primarily focus on machine health and equipment monitoring, this dataset integrates communication network performance indicators directly into the manufacturing analytics process. This enables the quantitative evaluation of communication-related factors such as latency variation and packet loss alongside production speed, operational errors, defect rates, and efficiency classifications. Such integration is essential for understanding the role of next-generation communication infrastructures within Industry 4.0 and Industry 5.0 environments. The dataset supports multiple analytical tasks, including network performance profiling, latency impact analysis, packet loss diagnostics, manufacturing efficiency evaluation, communication reliability assessment, operational risk identification, and KPI generation. By combining machine telemetry information with communication network measurements, the dataset provides a comprehensive foundation for investigating how network behavior affects manufacturing outcomes in smart factory environments. The complete dataset attributes used in this study are summarized in Table I. The dataset enables the development of analytical models and decision-support tools capable of identifying communication bottlenecks, evaluating operational risks, and supporting data-driven optimization of industrial communication infrastructures.

VI. METHODOLOGY

The proposed manufacturing analytics framework follows a structured analytical workflow consisting of multiple stages, including data preprocessing, network performance profiling, network quality segmentation, latency impact analysis, packet loss diagnostics, KPI generation, and dashboard visualization. The overall workflow of the proposed system is illustrated in Fig. 1.

Initially, the manufacturing dataset was loaded using Python and Pandas. Data preprocessing operations were performed to ensure consistency and reliability throughout the analytical process. These preprocessing steps included date-time conversion, validation of machine identifiers, handling missing values, verification of numerical attributes, and standardization of operational measurements. The processed dataset provided a reliable foundation for subsequent network and manufacturing performance analysis. Following data preparation, network performance profiling was conducted. The primary communication indicators analyzed in this study were Network Latency and Packet Loss. Statistical analysis was performed to evaluate communication behavior across the manufacturing environment. Distribution analysis, trend visualization, and summary statistics were used to identify stable and unstable communication conditions. These analyses enabled a compre-

TABLE I: Dataset Fields Description

Column Name	Description
Date	Calendar date on which the machine operational data was recorded.
Timestamp	Timestamp indicating the exact time of machine data collection and network measurement.
Machine_ID	Unique identifier assigned to each industrial machine operating within the smart factory environment.
Operation_Mode	Current operational mode of the machine, such as Normal or High-Load production conditions.
Temperature_C	Operating temperature of the machine measured in degrees Celsius.
Vibration_Hz	Machine vibration frequency measured in Hertz, representing mechanical operating behavior.
Power_Consumption_kW	Electrical power consumed by the machine during production operations, measured in kilowatts.
Network_Latency_ms	Communication latency experienced over the 6G industrial network, measured in milliseconds.
Packet_Loss_%	Percentage of communication packets lost during machine-to-machine and machine-to-system communication.
Quality_Control_Defect_Rate_%	Percentage of manufactured units identified as defective during quality-control inspection processes.
Production_Speed_units_per_hr	Number of production units generated by the machine per hour.
Predictive_Maintenance_Score	AI-generated score indicating the machine's maintenance readiness and overall health condition.
Error_Rate_%	Percentage of operational errors occurring during manufacturing processes.
Efficiency_Status	Target classification indicating overall manufacturing efficiency level (High, Medium, or Low).

hensive understanding of communication reliability within the smart factory infrastructure. After profiling network behavior, network quality segmentation was performed. Manufacturing records were categorized into different network quality levels based on latency and packet loss characteristics. Network conditions were classified into Low-Risk, Moderate-Risk, and High-Risk communication states to evaluate how varying communication quality affects manufacturing performance. This segmentation enabled comparative analysis between stable and degraded network environments. Subsequently, latency impact analysis was conducted to investigate the relationship between communication delay and manufacturing performance. Production speed measurements were analyzed across different latency ranges to identify performance sensitivity zones and communication thresholds associated with production degradation. Correlation analysis was performed to evaluate the influence of increasing latency on manufacturing efficiency classifications and operational throughput. Packet loss diagnostics were then performed to evaluate communication reliability effects on manufacturing quality and operational stability. The relationship between packet loss, operational error rates, and quality-control defect rates was analyzed using comparative statistical techniques and visual analytics. This analysis enabled the identification of communication reliability thresholds beyond which manufacturing performance begins to deteriorate significantly. To further investigate operational behavior, interaction analysis was conducted across different machine operation modes. Manufacturing records operating under Normal and High-Load conditions were compared to determine whether network degradation affects operational modes differently. This analysis provided insights into which production conditions are most sensitive to communication instability and operational disruptions. Several manufacturing-specific Key Performance Indicators (KPIs) were subsequently developed to quantify communication-related manufacturing

performance. These indicators included Network Stability Index, Latency Sensitivity Score, Packet Loss Impact Ratio, and Network-Efficiency Correlation metrics. The generated KPIs provide a standardized framework for evaluating communication reliability and manufacturing effectiveness within smart factory environments. Finally, an interactive Streamlit dashboard was developed using Plotly visualizations to provide comprehensive analytical insights. The dashboard integrates multiple modules including Network Performance Overview, Network versus Efficiency Analysis, Quality and Error Impact Diagnostics, and 6G Optimization Insights. Interactive filtering capabilities allow users to explore network conditions, operational modes, efficiency classifications, and temporal trends. The resulting platform provides a centralized decision-support environment for monitoring communication performance, identifying manufacturing risks, and supporting data-driven optimization of industrial communication infrastructures.

VII. KEY PERFORMANCE INDICATORS

To quantify the impact of communication network performance on manufacturing operations, several manufacturing-specific Key Performance Indicators (KPIs) were developed. These indicators provide a standardized framework for evaluating network reliability, operational efficiency, production stability, and communication-driven manufacturing risks within 6G-enabled smart factory environments.

A. Network Stability Index

The Network Stability Index (NSI) provides a combined measure of communication reliability by integrating network latency and packet loss behavior into a single performance indicator. The objective of this KPI is to evaluate overall network health and identify communication conditions that

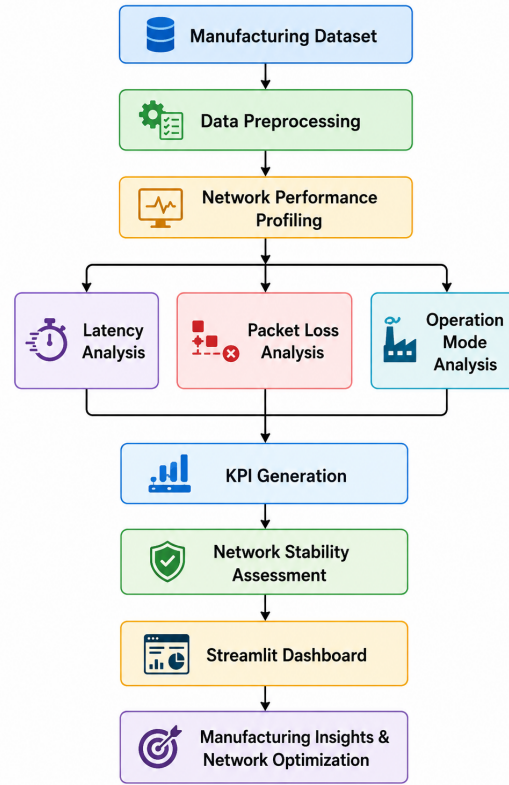


Fig. 1: Workflow of the proposed 6G network performance and manufacturing efficiency analysis framework.

may negatively affect manufacturing operations. Higher Network Stability Index values indicate stable communication environments characterized by low latency and minimal packet loss, while lower values indicate communication instability and increased operational risk. The Network Stability Index serves as a foundational indicator for assessing the readiness of industrial communication infrastructures supporting real-time manufacturing processes.

B. Latency Sensitivity Score

Latency Sensitivity Score measures the degree to which manufacturing performance changes as communication latency increases. This KPI evaluates the relationship between network delay and production efficiency by analyzing variations in production speed and efficiency classifications across different latency ranges. A higher Latency Sensitivity Score indicates that manufacturing operations are highly dependent on low-latency communication and may experience significant performance degradation when communication delays increase. The metric enables organizations to identify latency thresholds beyond which operational efficiency begins to decline.

C. Packet Loss Impact Ratio

Packet Loss Impact Ratio quantifies the influence of communication reliability on manufacturing performance. The metric evaluates how increases in packet loss affect operational error rates, product defect rates, and overall manufacturing

efficiency. Higher Packet Loss Impact Ratio values indicate that communication reliability issues significantly contribute to production degradation and operational instability. This KPI assists manufacturing organizations in identifying communication reliability thresholds that require network optimization and infrastructure improvement.

D. Network-Efficiency Correlation Score

Network-Efficiency Correlation Score measures the strength of the relationship between communication network conditions and manufacturing efficiency classifications. The metric evaluates how variations in latency and packet loss correspond to changes in efficiency status levels such as High, Medium, and Low efficiency categories. Strong correlation values indicate that communication network behavior serves as a major driver of manufacturing performance outcomes. This KPI provides quantitative evidence supporting communication-focused optimization strategies within smart factory environments.

E. Production Performance Degradation Index

Production Performance Degradation Index measures the reduction in manufacturing throughput associated with deteriorating network conditions. The indicator evaluates production speed behavior under different latency and packet loss scenarios to identify communication conditions responsible for operational slowdowns. The metric provides a standardized

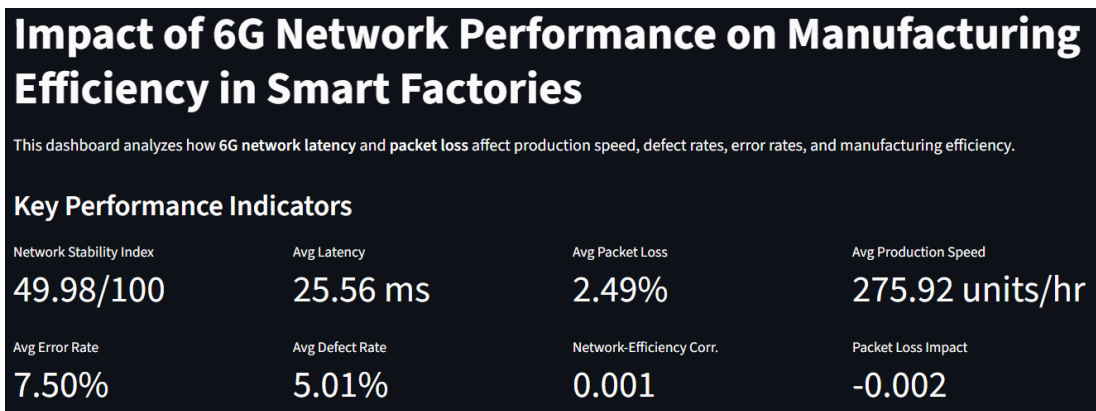


Fig. 2: Overview dashboard showing the main manufacturing and network performance indicators, including Network Stability Index, average latency, packet loss, production speed, error rate, defect rate, network-efficiency correlation, and packet loss impact.

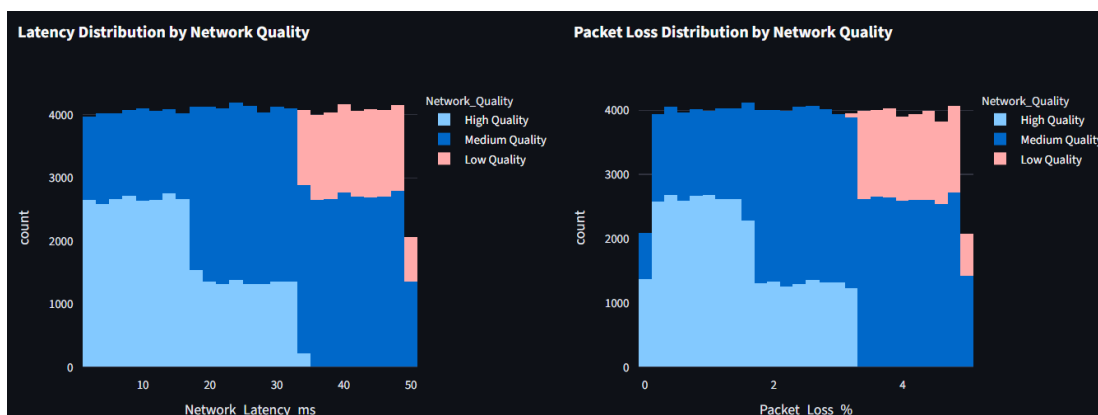


Fig. 3: Network performance overview showing latency and packet loss distributions across high-quality, medium-quality, and low-quality network conditions.

method for quantifying production losses attributable to communication instability. This KPI supports network investment decisions by demonstrating the operational consequences of degraded communication performance.

F. Quality Risk Indicator

Quality Risk Indicator evaluates the relationship between communication network degradation and product quality outcomes. The metric analyzes defect-rate behavior across varying network conditions to determine whether communication instability contributes to manufacturing quality issues. Elevated Quality Risk Indicator values suggest that latency and packet loss may negatively affect process synchronization, quality-control activities, and production consistency. This KPI enables manufacturers to proactively identify communication-related quality risks before they affect large-scale production operations.

G. Operational Reliability Score

Operational Reliability Score measures the ability of manufacturing systems to maintain stable performance under varying communication conditions. The metric combines infor-

mation from production speed, error rates, defect rates, and efficiency classifications to provide an overall assessment of manufacturing resilience. Higher reliability scores indicate that production systems remain stable despite communication fluctuations, while lower scores indicate increased sensitivity to network instability. This KPI provides a holistic view of manufacturing robustness and operational continuity within 6G-enabled industrial environments.

VIII. RESULTS AND DASHBOARD ANALYSIS

The developed Streamlit dashboard provides centralized visibility into the relationship between 6G network performance and manufacturing efficiency. Multiple analytical modules were created to evaluate latency behavior, packet loss impact, production speed variation, quality-control defect patterns, operational error rates, and network-driven efficiency changes. The dashboard includes interactive filters for network quality, efficiency status, operation mode, and date range, enabling detailed exploration of manufacturing performance under different communication conditions.

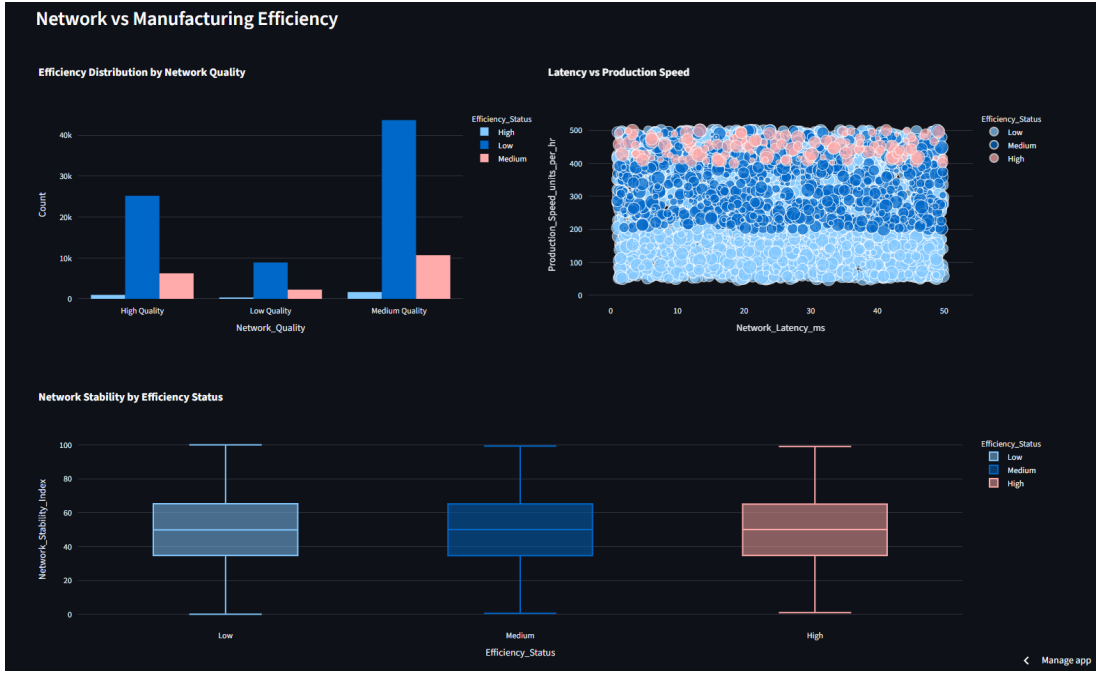


Fig. 4: Network versus manufacturing efficiency analysis showing efficiency distribution across network quality levels, latency versus production speed behavior, and network stability characteristics under different efficiency classifications.

A. Dataset Overview

Fig. 2 presents the main dashboard overview generated from the manufacturing dataset. The dashboard summarizes key performance indicators, including Network Stability Index, average latency, average packet loss, average production speed, average error rate, average defect rate, network-efficiency correlation, and packet loss impact. The overview provides a high-level understanding of how communication behavior and manufacturing performance are jointly monitored in the proposed system.

B. Network Performance Overview

Fig. 3 illustrates the network performance overview generated from the dashboard. The visualization presents latency distribution and packet loss distribution across different network quality levels. This analysis helps distinguish stable communication periods from unstable network conditions and provides the foundation for evaluating communication-related manufacturing risks.

C. Network versus Efficiency Analysis

Fig. 4 presents the relationship between network quality and manufacturing efficiency. The figure includes efficiency distribution by network quality, latency versus production speed behavior, and network stability variation across efficiency classes. The results help determine whether degraded communication conditions are associated with lower efficiency classifications and reduced production performance.

D. Latency Impact Diagnostics

Fig. 5 presents the latency impact diagnostics generated from the dashboard. The analysis evaluates how production speed varies with increasing network latency and helps identify latency-sensitive manufacturing conditions. This diagnostic view supports the identification of communication delay ranges that may contribute to production slowdowns.

E. Packet Loss Impact Diagnostics

Fig. 6 shows the packet loss impact diagnostics generated from the quality and error impact panel. The visualization examines the relationship between packet loss, operational error rate, and quality-control defect rate. The results demonstrate how communication reliability affects production stability and manufacturing quality outcomes.

F. 6G Optimization Insights

Fig. 7 presents the optimization insights generated from latency tolerance benchmarks, packet loss risk zones, and network stability indicators. These insights support decision-making for improving communication reliability, reducing operational risk, and optimizing 6G-enabled smart factory infrastructure.

IX. SYSTEM ARCHITECTURE

The proposed manufacturing analytics framework follows a modular architecture designed to transform raw manufacturing and communication data into actionable operational intelligence. The system integrates data preprocessing, network performance profiling, latency diagnostics, packet loss analysis, KPI generation, network stability assessment, and dashboard

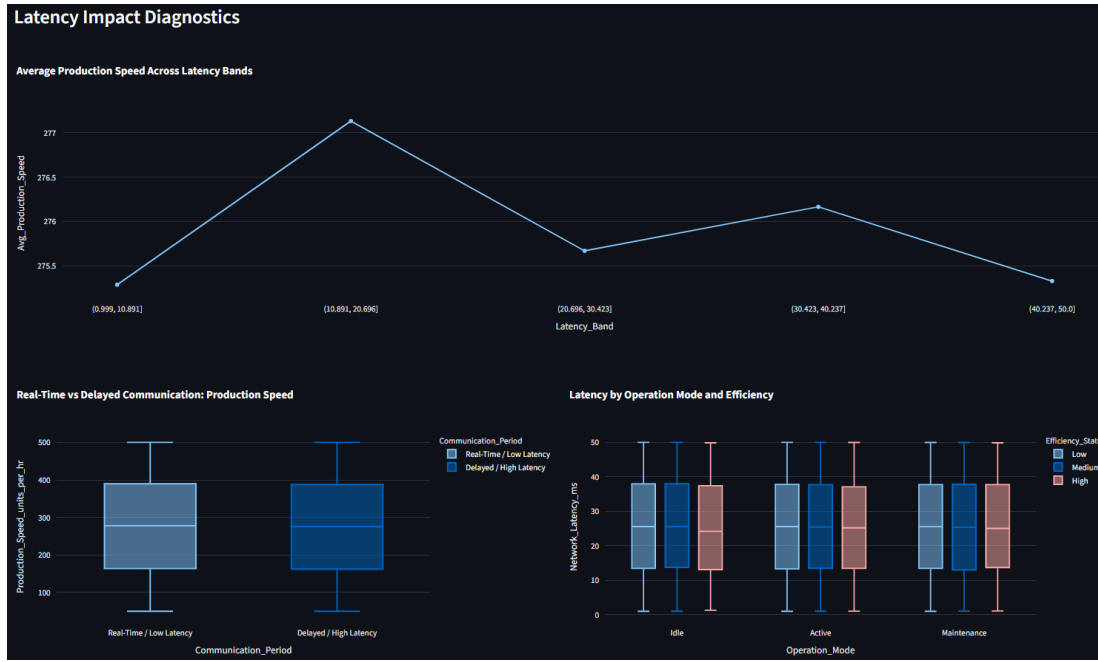


Fig. 5: Latency impact diagnostics showing the relationship between network latency and production speed, along with latency-sensitive manufacturing behavior.

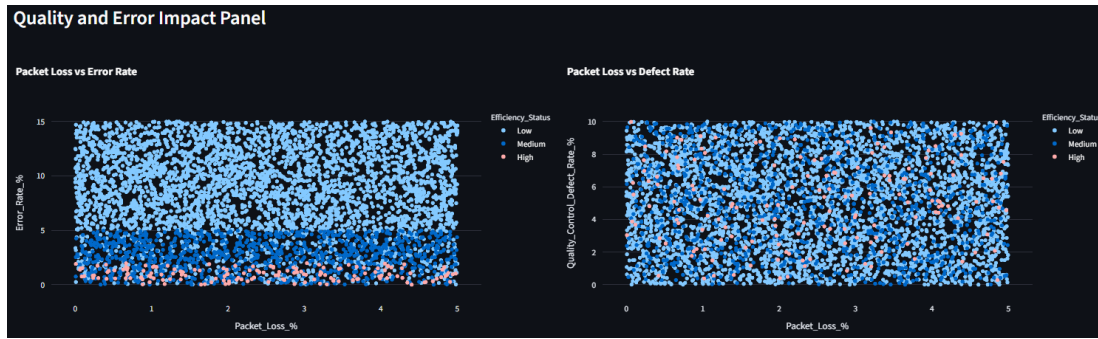


Fig. 6: Packet loss impact diagnostics showing the relationship between packet loss, operational error rate, and quality-control defect rate.

visualization into a unified analytical environment. The overall architecture of the proposed framework is illustrated in Fig. 8. The architecture begins with the Manufacturing Dataset, which serves as the primary source of machine telemetry, communication performance measurements, production metrics, quality-control indicators, and operational efficiency classifications. During the preprocessing stage, data cleaning, validation, normalization, missing-value handling, and feature preparation are performed to ensure analytical consistency and reliability. The processed data is subsequently analyzed through multiple specialized analytical modules. The Network Performance Profiling module evaluates latency distributions, packet loss behavior, and communication quality trends. The Latency Analysis module investigates the relationship between communication delay and manufacturing throughput, while the Packet Loss Analysis module examines the impact of communication reliability on defect rates and operational

errors. The Operation Mode Analysis module evaluates how different machine operating conditions respond to varying network environments. The outputs generated by these analytical modules are consolidated within the KPI Generation layer, where manufacturing-specific performance indicators are calculated. These indicators include the Network Stability Index, Latency Sensitivity Score, Packet Loss Impact Ratio, Network-Efficiency Correlation Score, Production Performance Degradation Index, Quality Risk Indicator, and Operational Reliability Score. The generated KPIs are subsequently utilized by the Network Stability Assessment engine to identify communication risk zones, evaluate manufacturing resilience, and quantify the overall impact of network conditions on operational performance. This layer provides decision-support capabilities for identifying communication bottlenecks and prioritizing network optimization efforts. Finally, all analytical results are integrated into an interactive Streamlit Dashboard

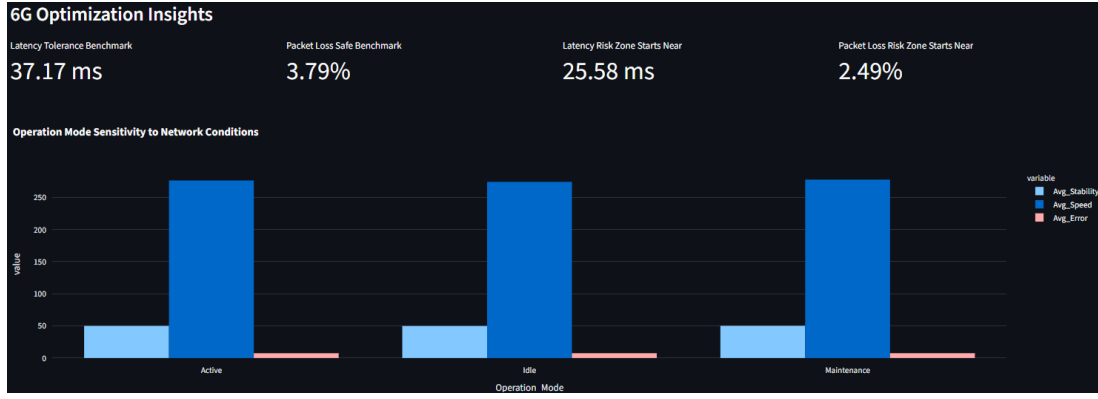


Fig. 7: 6G optimization insights showing latency tolerance benchmarks, packet loss risk zones, and network stability indicators for smart factory optimization.

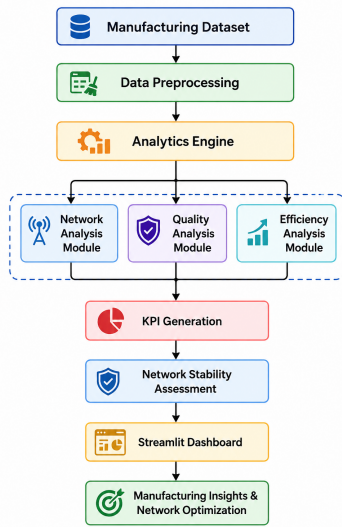


Fig. 8: System architecture of the proposed 6G network performance and manufacturing efficiency analysis framework.

that provides visual exploration of network performance, manufacturing efficiency behavior, quality-control outcomes, operational risks, and optimization opportunities. The resulting insights support strategic decision-making related to communication infrastructure planning, manufacturing optimization, operational reliability improvement, and next-generation smart factory deployment.

X. KEY INSIGHTS

The analysis performed using the proposed framework revealed several important relationships between communication network performance and manufacturing efficiency within 6G-enabled smart factory environments.

A. Communication Stability Directly Influences Manufacturing Efficiency

The dashboard analysis demonstrated that network performance serves as an independent factor affecting manufac-

turing efficiency. Variations in communication quality were associated with observable differences in production speed, operational reliability, and efficiency classifications. Stable communication environments consistently supported improved manufacturing performance, while degraded network conditions increased operational uncertainty and reduced efficiency consistency.

B. Network Latency Affects Production Throughput

The latency diagnostics revealed that increasing communication delay negatively influences production responsiveness and operational throughput. Although manufacturing systems remained functional under varying latency conditions, elevated communication delays were associated with reduced production speed and less stable operational behavior. This finding highlights the importance of maintaining ultra-low latency communication within smart manufacturing environments.

C. Packet Loss Contributes to Operational Risk

Packet loss analysis indicated that communication reliability significantly affects manufacturing quality and operational stability. Increased packet loss was associated with elevated error rates and higher quality-control defect rates. The results suggest that communication reliability should be monitored alongside traditional machine-health indicators to ensure consistent manufacturing performance.

D. Network Stability Serves as a Leading Performance Indicator

The Network Stability Index provided a consolidated measure of communication health and proved effective for identifying manufacturing risk conditions. Lower stability levels were frequently associated with increased operational variability, suggesting that network stability metrics can serve as early-warning indicators for potential production disruptions.

E. Manufacturing Efficiency Cannot Be Evaluated Using Machine Data Alone

A major finding of this study is that machine telemetry alone does not provide a complete representation of manufacturing

performance. Communication-related variables such as latency and packet loss contribute directly to operational outcomes and should be incorporated into future manufacturing analytics frameworks. The results demonstrate the importance of adopting a connectivity-first perspective when evaluating industrial efficiency.

F. Integrated Analytics Improve Decision-Making

The combination of network analytics, manufacturing performance evaluation, KPI generation, and interactive dashboard visualization provides a comprehensive decision-support framework for industrial stakeholders. The proposed approach enables organizations to identify communication bottlenecks, prioritize network optimization efforts, and improve overall manufacturing resilience through data-driven decision-making.

XI. RECOMMENDATIONS

Based on the analytical findings obtained from the proposed framework, several recommendations can be made to improve manufacturing performance and communication reliability within 6G-enabled smart factory environments. First, manufacturing organizations should continuously monitor network latency and packet loss alongside traditional machine-health indicators. Communication-related performance metrics should be incorporated into operational dashboards to enable early detection of network-induced production inefficiencies. Second, manufacturers should establish latency tolerance benchmarks for critical production processes. Identifying acceptable communication delay thresholds can help prevent production slowdowns and support proactive network management strategies. Third, packet loss monitoring mechanisms should be integrated into industrial communication infrastructures. Since communication reliability directly affects operational error rates and product quality, organizations should prioritize network optimization efforts that minimize packet loss during production activities. Fourth, network stability indicators should be incorporated into predictive manufacturing and operational risk assessment systems. The Network Stability Index proposed in this study can serve as an early-warning mechanism for identifying communication conditions that may negatively affect manufacturing performance. Fifth, organizations should adopt integrated analytics frameworks that combine machine telemetry, production metrics, quality-control indicators, and network performance measurements within a unified decision-support environment. Such an approach provides a more comprehensive understanding of manufacturing operations than traditional machine-centric monitoring systems.

Finally, industrial stakeholders should consider communication infrastructure performance as a strategic operational asset rather than merely a supporting technology. Investments in reliable, low-latency communication networks can directly contribute to improved manufacturing efficiency, reduced operational risk, and enhanced production quality.

XII. FUTURE WORK

Several opportunities exist for extending the proposed framework and improving its analytical capabilities. Future research can incorporate real-world industrial communication data collected from operational smart factories to validate the findings under practical manufacturing conditions. Additional network performance indicators such as jitter, bandwidth utilization, throughput variability, and communication congestion can also be integrated into the analytical framework. Machine learning and artificial intelligence techniques may be utilized to develop predictive models capable of forecasting manufacturing efficiency degradation based on communication network behavior. Such models could enable proactive network management and automated optimization strategies. Future studies may also investigate the impact of network slicing, edge computing architectures, and intelligent resource allocation mechanisms within 6G-enabled industrial environments. These technologies are expected to play a significant role in supporting next-generation smart manufacturing systems. Another promising direction involves the integration of digital twin technologies with communication performance analytics. Combining real-time network monitoring with digital twin simulations could enable dynamic evaluation of manufacturing scenarios and communication risk assessment. Finally, future work can expand the proposed framework to support multi-factory environments, large-scale industrial ecosystems, and Industry 5.0 applications involving collaborative robots, autonomous systems, and human-machine interaction platforms.

XIII. CONCLUSION

This paper presented an analytical framework for evaluating the impact of 6G network performance on manufacturing efficiency within smart factory environments. The study investigated how communication-related factors, including network latency and packet loss, influence production speed, operational error rates, product quality, and overall manufacturing efficiency. A comprehensive manufacturing analytics framework was developed that integrates network performance profiling, latency diagnostics, packet loss analysis, manufacturing efficiency evaluation, KPI generation, and interactive dashboard visualization. Several manufacturing-specific Key Performance Indicators were introduced, including the Network Stability Index, Latency Sensitivity Score, Packet Loss Impact Ratio, Network-Efficiency Correlation Score, Production Performance Degradation Index, Quality Risk Indicator, and Operational Reliability Score. The results demonstrated that communication network performance serves as a critical and independent driver of manufacturing outcomes. Network instability was shown to influence operational reliability, production throughput, manufacturing efficiency, and quality-control performance. These findings highlight the importance of incorporating communication analytics into modern manufacturing decision-support systems. An interactive Streamlit dashboard was developed to provide real-time visibility into network behavior, manufacturing performance trends, communication risk zones, and optimization opportunities. The

proposed framework enables industrial stakeholders to identify communication bottlenecks, improve operational resilience, support data-driven decision-making, and optimize 6G-enabled manufacturing infrastructures. Overall, this study demonstrates that manufacturing efficiency can no longer be evaluated solely through machine-health and process-performance indicators. As smart factories continue to evolve toward Industry 5.0 and beyond, communication network performance will play an increasingly important role in determining operational success, production reliability, and industrial competitiveness.

REFERENCES

- [1] K. Zhou, T. Liu, and L. Zhou, "Industry 4.0: Towards Future Industrial Opportunities and Challenges," *12th International Conference on Fuzzy Systems and Knowledge Discovery*, pp. 2147–2152, 2015.
- [2] H. Lasi, P. Fettke, H.-G. Kemper, T. Feld, and M. Hoffmann, "Industry 4.0," *Business & Information Systems Engineering*, vol. 6, no. 4, pp. 239–242, 2014.
- [3] L. D. Xu, W. He, and S. Li, "Internet of Things in Industries: A Survey," *IEEE Transactions on Industrial Informatics*, vol. 10, no. 4, pp. 2233–2243, 2014.
- [4] Y. Lu, "Industry 4.0: A Survey on Technologies, Applications and Open Research Issues," *Journal of Industrial Information Integration*, vol. 6, pp. 1–10, 2017.
- [5] W. Saad, M. Bennis, and M. Chen, "A Vision of 6G Wireless Systems: Applications, Trends, Technologies, and Open Research Problems," *IEEE Network*, vol. 34, no. 3, pp. 134–142, 2020.
- [6] F. Tariq, M. R. Khandaker, K.-K. Wong, M. A. Imran, M. Bennis, and M. Debbah, "A Speculative Study on 6G," *IEEE Wireless Communications*, vol. 27, no. 4, pp. 118–125, 2020.
- [7] J. Lee, H.-A. Kao, and S. Yang, "Service Innovation and Smart Analytics for Industry 4.0 and Big Data Environment," *Procedia CIRP*, vol. 16, pp. 3–8, 2014.
- [8] T. P. Carvalho, F. A. Soares, R. Vita, R. P. Francisco, J. P. Basto, and S. G. Alcalá, "A Systematic Literature Review of Machine Learning Methods Applied to Predictive Maintenance," *Computers & Industrial Engineering*, vol. 137, 2019.
- [9] L. Monostori et al., "Cyber-Physical Systems in Manufacturing," *CIRP Annals*, vol. 65, no. 2, pp. 621–641, 2016.
- [10] F. Tao, Q. Qi, A. Liu, and A. Kusiak, "Data-Driven Smart Manufacturing," *Journal of Manufacturing Systems*, vol. 48, pp. 157–169, 2018.

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